

13F Manager Factor Decomposition and Factor Crowding

Implementation memo, literature review, and research roadmap

Research memo

16 August 2026

Executive recommendation

The next missing research block in the current 13F project is a **holdings-based decomposition of each manager's long-equity book into classical systematic factors**, followed by a separate layer that measures **factor positioning, factor crowding, and factor flow across managers**.

This is not the same thing as the factor work already documented in CONCLUSOES_13F.pdf.

The current project already contains:

- factor-adjusted performance tests of 13F signals (CAPM -> FF3 -> FF5 -> FF6 -> short-term reversal);
- latent crowding/state methods such as NMF, PCA/sig_share, network centrality and Bonacich-type measures;
- a validated manager-level implicit flow instrument;
- predictive flow signals such as new_conviction and distress_mom;
- an explicit methodological conclusion that **state has no directional arrow, while flow does**.

What is not documented as a completed block is:

13F manager portfolio → classical factor exposure vector → factor positioning/crowding across managers

The recommended implementation is therefore **two-stage**:

1. **Descriptive/risk layer:** build a manager-by-factor exposure matrix from holdings.
2. **Predictive layer:** convert changes in those exposures, manager funding shocks, and crowding vulnerability into directional factor-flow signals.

The first stage is relatively easy and should be implemented first. The second stage is where the best potential alpha is likely to live, given the evidence already found in the project.

A concise roadmap is:

Sprint 1: market-data-only factor decomposition -> **Sprint 2:** factor crowding dashboard -> **Sprint 3:** factor-flow pressure -> **Sprint 4:** fundamentals-based factors -> **Sprint 5:** factor stress matrix / eigenmodes.

1. What exactly are we trying to measure?

At each 13F snapshot, suppose there are:

- M_t managers;
- N_t stocks;
- K factors.

Define the manager-stock holdings matrix:

$$W_t \in \mathbb{R}^{M_t \times N_t},$$

where $w_{mi,t}$ is the weight of stock i in manager m 's reported 13F long-equity book.

For the cleanest first version:

$$w_{mi,t} = \frac{\text{market value}_{mi,t}}{\sum_j \text{market value}_{mj,t}}.$$

Thus each manager's reported equity book satisfies approximately:

$$\sum_i w_{mi,t} = 1.$$

Now define a stock-factor exposure matrix:

$$X_t \in \mathbb{R}^{N_t \times K},$$

where $x_{ik,t}$ is stock i 's exposure or characteristic score for factor k .

The core object is then simply:

$$E_t = W_t X_t$$

with

$$E_t \in \mathbb{R}^{M_t \times K}.$$

For a manager m and factor k :

$$E_{mk,t} = \sum_i w_{mi,t} x_{ik,t}$$

is the manager's holdings-based exposure to that factor.

Important: factor exposures do not sum to one

These are not capital weights. There is no reason for

$$\sum_k E_{mk,t} = 1.$$

A manager can simultaneously be:

- +1.3 standard deviations in Momentum;
- +0.8 in Quality;
- -0.5 in Size (i.e. tilted to large caps, depending on sign convention);
- +0.4 in Low Volatility.

The factors are coordinates in a feature/risk space, not mutually exclusive portfolio buckets.

Sector dummy exposures are a special case: if every stock belongs to exactly one sector and the book contains only classified common stocks, sector weights will approximately sum to one. Style factors generally will not.

2. Three different meanings of “factor exposure” - do not mix them

A critical design choice is to distinguish **characteristic tilts** from **return betas**.

2.1 Holdings-based characteristic exposure - recommended headline

For each stock, construct a cross-sectional characteristic score:

$$x_{ik,t} = z(\text{characteristic}_{ik,t}).$$

Examples:

- Size: negative standardized log market capitalization if positive means “small”;
- Momentum: standardized prior 12-month return excluding the most recent month;
- Value: standardized book-to-market or earnings yield;
- Profitability: standardized operating profitability;
- Investment: negative standardized asset growth if positive means “conservative”;
- Low Volatility: negative standardized residual volatility or beta;
- Liquidity: a deliberately signed ADV/Amihud measure.

Then:

$$E_{mk,t}^{char} = \sum_i w_{mi,t} x_{ik,t}.$$

This is the most natural object for 13F positioning because it can be computed from a **single holdings snapshot** and can change quickly when the manager rotates the book.

2.2 Holdings-implied return-beta exposure - useful robustness test

Estimate stock-level factor betas from daily returns:

$$r_{i,d} - r_{f,d} = \alpha_i + \sum_k \beta_{ik,t} F_{k,d} + \epsilon_{i,d},$$

using only information available up to date t .

Then aggregate through holdings:

$$E_{mk,t}^{\beta} = \sum_i w_{mi,t} \beta_{ik,t}.$$

This answers a different question:

What systematic return risk would this reported long-equity book likely carry?

It is useful, but it is more estimation-heavy and can be noisy when stock betas are unstable.

2.3 Manager return regression - diagnostic, not primary

Construct a synthetic return series for the manager’s disclosed equity book and regress it on factor returns:

$$r_{m,d}^{book} - r_{f,d} = \alpha_m + \sum_k \beta_{mk} F_{k,d} + \epsilon_{m,d}.$$

This is useful for validation, but it is less suitable as the main positioning measure because the beta needs a return history and implicitly smooths exposures through time.

Recommendation

Use:

$$E_{mk,t}^{char}$$

as the primary factor-positioning coordinate.

Use holdings-implied and return-based betas as **robustness and validation layers**, not as the headline definition.

This preference is aligned with holdings-based style research: portfolio characteristics are directly interpretable from holdings and can be measured at a specific snapshot, while return-based loadings require a window and impose more persistence on the estimated exposure.

3. Which classical factors should be implemented?

The implementation should be staged according to data quality.

3.1 Tier 0 - implement immediately with 13F + market data

These can be built without point-in-time accounting data.

Factor	Stock-level variable	Suggested sign	Comment
Market beta	rolling CAPM beta	positive = high beta	risk exposure, not a style premium by itself
Size	$\log(ME)$	positive = small via $-z(\log ME)$	canonical SMB direction
Momentum	return from $t - 12m$ to $t - 1m$	positive = winner	canonical 12-1 style
Low Vol / Defensive	residual vol or beta	positive = low risk	keep separate from beta initially
Liquidity / Illiquidity	$\log ADV$, turnover, Amihud	define explicitly	important because 13F mechanically loads on liquid large caps
Short-term reversal	prior 1-month return	positive or negative by convention	useful control
Sectors	one-hot industry/sector dummies	weight share	necessary to distinguish sector bets from style bets

Minimal recommended set

If speed matters, the first production table should contain only:

SIZE, MOM, BETA, LOWVOL, LIQUIDITY, SECTORS

This is enough to answer a large part of the economic question without introducing stale fundamentals.

3.2 Tier 1 - add canonical accounting characteristics once PIT data are clean

Add:

- Value: book-to-market, earnings yield or a robust composite;
- Profitability / Quality: operating profitability, gross profitability or a quality composite;
- Investment: asset growth / conservative investment;
- possibly leverage, earnings stability and payout as additional quality dimensions.

For strict Fama-French language:

- HML is linked to high versus low book-to-market;
- RMW to robust versus weak profitability;
- CMA to conservative versus aggressive investment;
- momentum uses prior 2-12 month returns in the Kenneth French construction.

Do **not** use current Yahoo fundamentals in a historical backtest unless their point-in-time availability has been verified. A clean free-data extension is SEC EDGAR/XBRL, where filing timestamps make PIT accounting variables possible.

3.3 Tier 2 - non-classical but desk-relevant exposures

After the core classical factors work, extend to:

- Growth;
- duration / long-duration equity proxy;
- dividend yield;
- profitability stability;
- leverage;
- idiosyncratic volatility;
- crowding-liquidity interaction;
- AI/technology thematic baskets if there is a clearly defined rule.

These should be treated as extensions, not substitutes for the canonical factor layer.

4. How should stock characteristics be standardized?

This matters more than it looks.

A raw weighted average such as

$$\sum_i w_i \log(ME_i)$$

will be dominated by scale and will be difficult to compare across dates.

The preferred procedure at every factor date is:

1. choose an eligible stock universe;
2. winsorize the raw characteristic, e.g. at 1%/99%;
3. transform it if needed ($\log(\text{ME})$, $\log(\text{ADV})$, etc.);
4. cross-sectionally standardize;
5. preserve a fixed sign convention.

For example:

$$z_{i,k,t} = \frac{x_{i,k,t} - \mu_{k,t}}{\sigma_{k,t}}.$$

A robust version can replace mean/std with median/MAD or use cross-sectional ranks mapped to Gaussian scores.

Global versus sector-neutral scores

Compute **both**.

Raw/global exposure

$$E_{mk,t}^{raw} = \sum_i w_{mi,t} z_{ik,t}^{global}.$$

This describes what the manager actually owns economically.

Sector-neutral exposure

First residualize the stock characteristic on sector dummies:

$$x_{ik,t} = \alpha_{s(i),k,t} + u_{ik,t},$$

then standardize $u_{ik,t}$ and aggregate:

$$E_{mk,t}^{SN} = \sum_i w_{mi,t} z(u_{ik,t}).$$

This asks:

Within the sectors the manager chose, is the manager still tilted to Momentum, Value, Size, etc.?

Do not choose one and discard the other. The raw and sector-neutral versions answer different questions.

5. Benchmark-relative exposure is preferable to raw exposure

A 13F long-equity book has unavoidable structural tilts: it will tend to contain U.S.-listed, liquid, larger stocks.

Therefore a useful manager exposure is the tilt relative to a benchmark portfolio b_t :

$$\widetilde{E}_{mk,t} = \sum_i (w_{mi,t} - b_{i,t}) x_{ik,t}$$

Possible benchmarks:

1. value-weighted eligible 13F universe;
2. S&P 500 / Russell-type benchmark if manager mandate is known;
3. manager-specific historical benchmark inferred from its own holdings;
4. equal-weighted universe as a robustness check.

Recommended default

Use the **value-weighted eligible 13F stock universe** as the baseline:

$$b_{i,t} = \frac{ME_{i,t}}{\sum_j ME_{j,t}}.$$

Then the aggregate market portfolio has approximately zero style tilt by construction.

This is especially important for Size and Liquidity. Otherwise the finding that most managers own large/liquid stocks can be partly mechanical rather than informative.

6. The manager-by-factor matrix: the main new data product

For each PIT 13F snapshot, store:

$$E_t = \begin{bmatrix} E_{1,1,t} & \cdots & E_{1,K,t} \\ \vdots & \ddots & \vdots \\ E_{M,1,t} & \cdots & E_{M,K,t} \end{bmatrix}.$$

A practical table schema is:

```
factor_exposure_manager
-----
decision_date
holdings_quarter
filer_id
factor_name
exposure_raw
exposure_sector_neutral
exposure_active_vs_market
factor_percentile_among_managers
equity_13f_aum
coverage_pct
n_positions
```

Coverage quality

Every manager exposure should carry a coverage field:

$$coverage_{m,t} = \sum_{i \in \text{mapped factor universe}} w_{mi,t}.$$

Do not compare a manager with 98% mapped holdings to one with 45% mapped holdings without flagging the difference.

A reasonable first production filter is:

$$coverage_{m,t} \geq 80\%.$$

Test 70%, 80%, and 90% as robustness bands rather than optimizing the threshold.

7. Factor positioning: how is the institutional market positioned?

Once E_t exists, several market-level statistics become trivial.

Let $a_{m,t}$ be a manager aggregation weight.

The project evidence argues against skill-weighting or selecting only “smart money.” Therefore the default should use the **full eligible manager set** and report multiple mechanical weighting schemes rather than choosing managers by ex-post skill.

Recommended manager weights:

- equal-weight managers;
- 13F equity AUM weight;
- capped/square-root AUM weight as a robustness compromise.

7.1 Aggregate factor position

$$P_{k,t} = \sum_m a_{m,t} E_{mk,t}$$

Interpretation:

What is the aggregate long-equity institutional tilt to factor k ?

7.2 Factor breadth

Define a threshold c , for example 0, +0.5, or the manager cross-sectional 70th percentile.

$$B_{k,t}^+ = \sum_m a_{m,t} \mathbf{1}(E_{mk,t} > c)$$

and

$$B_{k,t}^- = \sum_m a_{m,t} \mathbf{1}(E_{mk,t} < -c).$$

This distinguishes:

- a factor held moderately by almost everyone;
- a factor held extremely by a small group.

7.3 Dispersion

$$D_{k,t} = \sqrt{\sum_m a_{m,t} (E_{mk,t} - P_{k,t})^2}.$$

Low dispersion + high mean exposure means broad consensus.

High dispersion means polarization.

7.4 Tail crowding

A useful statistic is capital in the extreme positive tail:

$$TC_{k,t}^+ = \sum_m a_{m,t} \max(E_{mk,t} - q_{0.75,k,t}, 0).$$

Analogously for the negative tail.

7.5 Concentration of factor ownership across managers

For the positive side, define:

$$q_{mk,t}^+ = \frac{a_{m,t} E_{mk,t}^+}{\sum_j a_{j,t} E_{jk,t}^+}, \quad E^+ = \max(E, 0).$$

Then:

$$HHI_{k,t}^+ = \sum_m (q_{mk,t}^+)^2$$

Compute the negative side separately.

This distinguishes “everyone owns the factor” from “the factor exposure is concentrated in a few giant books.”

8. A factor crowding dashboard should not be a single number initially

The project has already found a key lesson: **crowding/state alone does not determine direction**.

Therefore do not start with:

$$Crowding_{k,t} > 0 \Rightarrow Short Factor_k.$$

Instead, build a dashboard with separate dimensions:

Position + Breadth + Concentration + Flow + Fragility

For each factor, report something like:

Momentum

Aggregate Position	+1.6 z
Manager Breadth	73%
Positive-tail HHI	0.08
Quarterly Exposure Flow	+0.9 z
Funding Fragility	+1.4 z

The first three are **state**.

The last two can create a **directional mechanism**.

That distinction should remain explicit in the research design.

9. The most important extension: factor exposure FLOW

Static factor position is descriptive. The more promising signal is the change in factor exposure.

A naive change is:

$$\Delta E_{mk,t} = E_{mk,t} - E_{mk,t-1}.$$

But this mixes:

- price drift;
- changes in stock characteristics;
- actual manager trading.

A cleaner decomposition is possible.

9.1 Exact sequential decomposition

Start from the previous-quarter book:

$$E_{mk}^{old} = \sum_i w_{mi,t-1} X_{ik,t-1}.$$

Step A - price drift only

Drift the old weights forward using stock returns, without manager trades:

$$\tilde{w}_{mi,t} = \frac{w_{mi,t-1}(1 + R_{i,t-1 \rightarrow t})}{\sum_j w_{mj,t-1}(1 + R_{j,t-1 \rightarrow t})}.$$

Then:

$$E_{mk,t}^{price} = \sum_i \tilde{w}_{mi,t} X_{ik,t-1}.$$

Step B - characteristic drift

Update factor characteristics while keeping the drifted old book:

$$E_{mk,t}^{char} = \sum_i \tilde{w}_{mi,t} X_{ik,t}.$$

Step C - actual new holdings

$$E_{mk,t}^{actual} = \sum_i w_{mi,t} X_{ik,t}.$$

Now the exposure change decomposes exactly as:

$$E_{mk,t}^{actual} - E_{mk,t}^{old} = \underbrace{(E_{mk,t}^{price} - E_{mk,t}^{old})}_{\text{price drift}} + \underbrace{(E_{mk,t}^{char} - E_{mk,t}^{price})}_{\text{characteristic drift}} + \underbrace{(E_{mk,t}^{actual} - E_{mk,t}^{char})}_{\text{manager trading}}.$$

The object of greatest interest is:

$$\boxed{Rotation_{mk,t} = E_{mk,t}^{actual} - E_{mk,t}^{char}}$$

because it isolates the factor rotation caused by actual portfolio changes rather than passive drift.

Why this is attractive in this project

The existing 13F work found that removing mechanical components often destroyed rather than improved the signal at the stock level. Therefore the decomposition should **measure all components**, not automatically residualize them away.

Report:

- price drift;
- characteristic drift;
- active rotation;
- total change.

Then test which component contains predictive content.

10. Aggregate factor-flow pressure

Convert manager-level factor rotation into an institutional factor demand measure.

A simple dollar-scaled version is:

$$Flow_{k,t}^{trade} = \sum_m AUM_{m,t-1}^{13F} Rotation_{mk,t}.$$

Standardize through time or divide by an estimate of factor capacity.

A practical first normalization is:

$$Pressure_{k,t}^{trade} = \frac{Flow_{k,t}^{trade}}{\sum_i |x_{ik,t}| ADV_{i,t}}$$

The denominator approximates the trading capacity of the long/short factor basket.

The interpretation becomes:

How many “factor-ADV units” of institutional demand were directed into this factor?

This is much closer to a tradable mechanism than a static crowding score.

11. Integrating the already validated implicit manager flow

The current project already has a manager-level implicit AUM-flow instrument validated against subsequent selling behavior.

Let:

$$f_{m,t}$$

be the estimated dollar inflow/outflow for manager m .

If a manager begins the period with factor exposure $E_{mk,t-1}$, a first-order funding-induced factor demand is:

$$Flow_{k,t}^{funding} = \sum_m f_{m,t} E_{mk,t-1}$$

This gives a direct bridge between the strongest validated instrument already in the project and the new factor-positioning layer.

Do not double count

Observed holding changes already reflect both funding and discretionary trading. Therefore keep two definitions separate:

1. **Observed factor rotation** from actual holdings changes;
2. **Predicted funding-induced factor demand** from the flow instrument.

Then study:

$$ObservedFlow_{k,t} = \alpha + \beta FundingFlow_{k,t} + \epsilon_{k,t}.$$

The residual is a useful candidate for discretionary factor rotation, but - consistent with the lessons already learned - it should be tested rather than assumed to be more informative.

12. Factor crowding as a risk conditioner, not automatically a directional signal

The empirical literature does not support a universal rule “crowded factor $\rightarrow$ imminent negative return.”

That is also consistent with the current project’s own NMF and network results.

A better hypothesis is interaction-based:

$$\boxed{\text{Directional Shock} \times \text{Crowding Vulnerability}}$$

For example:

$$UnwindRisk_{k,t} = (-FundingFlow_{k,t})^+ \times CrowdingState_{k,t}.$$

Possible crowding states include:

- extreme aggregate position;
- low manager dispersion;
- high tail concentration;
- high ownership overlap;
- high funding-flow covariance among factor holders.

The basic economic thesis becomes:

A factor being crowded is not enough. The factor becomes dangerous when the capital supporting that factor receives a common adverse shock.

13. Advanced mathematical extension: Factor Stress Matrix

This is the natural high-end extension once E_t exists.

Let manager percentage funding shocks be:

$$u_t \in \mathbb{R}^M,$$

and estimate a shrinkage covariance matrix:

$$\Sigma_{u,t} = Cov(u).$$

Define manager dollar factor exposures:

$$L_t = \text{diag}(AUM_t^{13F})E_t.$$

Then construct:

$$\boxed{\Gamma_t = L_t^\top \Sigma_{u,t} L_t}$$

where

$$\Gamma_t \in \mathbb{R}^{K \times K}.$$

Interpretation:

- Γ_{kk} : vulnerability of factor k to common manager funding shocks;
- Γ_{kl} : tendency for factors k and l to be hit by the same underlying capital shocks.

Eigendecompose:

$$\Gamma_t = Q_t \Lambda_t Q_t^\top.$$

The first eigenvector is the dominant **institutional factor stress mode**.

It may look economically like a combination such as:

$$q_1 \approx 0.7 \text{ MOM} + 0.4 \text{ SIZE} - 0.5 \text{ VALUE} + \dots$$

but the signs and weights are learned from the data rather than imposed.

Now project the current funding shock onto factor space:

$$g_t = L_t^\top u_t, \quad a_t = Q_t^\top g_t.$$

A large negative $a_{1,t}$ means the current flow shock is aligned with the most vulnerable institutional factor combination.

This construction is much more economically identified than a network centrality measure alone because it explicitly links:

who owns the factors + whose funding moves together + the current directional shock

14. A second advanced extension: factor rebalancing mismatch

Recent work by Cameron Peng and Chen Wang motivates another particularly relevant direction.

The core idea is that managers may have persistent preferences for styles such as Value or Momentum, while the characteristics of individual stocks drift through time. A stock can therefore become **misaligned with the factor preference of its current owners**, creating predictable future rebalancing pressure.

For each manager, estimate a persistent factor preference:

$$\bar{E}_{mk,t} = \text{slow-moving average of } E_{mk,s}, \quad s < t.$$

For a stock i , compute the factor preference of its ownership base:

$$OwnerPref_{ik,t} = \frac{\sum_m h_{mi,t} \bar{E}_{mk,t}}{\sum_m h_{mi,t}},$$

where $h_{mi,t}$ is a dollar holding or normalized ownership exposure.

Then compare the current stock characteristic $X_{ik,t}$ with the preference of its holders.

A stylized mismatch is:

$$Mismatch_{ik,t} = OwnerPref_{ik,t} \times (-\Delta X_{ik,t}),$$

with the precise sign depending on the factor.

Example:

- a manager persistently targets Momentum;
- a stock in the book was a strong winner;

- its momentum score subsequently collapses;
- the manager has not yet fully exited because holdings adjust with friction;
- future 13F trading may mechanically sell the now-misaligned stock.

This is a stock-level alpha extension of the manager-factor exposure matrix and should be a high-priority project after the basic factor decomposition works.

15. Point-in-time implementation: two clocks must be preserved

This project already has a major advantage: actual 13F filing timestamps.

Factor decomposition introduces a second clock:

1. **holdings reference date:** quarter-end;
2. **information availability date:** filing acceptance timestamp.

Keep two separate versions of exposure.

15.1 As-reported historical exposure

Use quarter-end holdings and quarter-end stock characteristics:

$$E_{mk,q}^{QE} = \sum_i w_{mi,q}^{QE} X_{ik,QE}.$$

This is the cleanest object for describing what the manager’s portfolio looked like at quarter-end.

15.2 Tradable current exposure

At decision date d , after the filing becomes public:

- take the disclosed shares;
- drift their market values using prices to d ;
- update market-data characteristics to d ;
- do not use information from managers that had not yet filed by d .

Then:

$$E_{mk,d}^{PIT} = \sum_i w_{mi,d}^{drift} X_{ik,d}.$$

This answers:

Given what is public right now, what factor exposure is implied by the manager’s last disclosed long-equity book?

Both objects are useful and should never be silently mixed.

16. 13F limitations specific to factor decomposition

This section is essential because the label “manager factor exposure” can otherwise be misleading.

16.1 13F is not the manager's full portfolio

The SEC 13F framework primarily covers reportable U.S. exchange-traded equities and certain other Section 13(f) securities. It does not reveal the complete economic portfolio of a hedge fund.

Missing or incomplete dimensions can include:

- most short positions;
- swaps and many derivatives;
- bonds and other non-13F assets;
- foreign securities outside the reportable set;
- cash;
- confidential positions during the confidentiality period.

Therefore the correct label is:

13F long-equity factor exposure

not simply “fund factor exposure.”

16.2 Hedge funds can be especially misleading

A hedge fund can report a large long Momentum tilt in 13F while hedging or reversing that exposure through shorts, futures or swaps outside the filing.

This does not make the signal useless. It changes its interpretation:

The measure describes the factor content and potential flow pressure of the **public long-equity leg**.

For crowding and disclosure-driven price pressure, this can still be precisely the relevant object.

16.3 Options

For the first version, exclude option rows rather than pretending the reported notional is a delta-equivalent stock position.

Add options only when a consistent option-type and delta treatment is available.

17. Validation before testing alpha

Before asking whether factor crowding predicts returns, prove that the exposure layer measures what it claims.

Test 1 - synthetic known portfolios

Create artificial books:

- top-decile Momentum stocks;
- bottom-decile Momentum stocks;
- small-cap basket;
- low-beta basket.

Verify that the manager-factor engine assigns the expected signs and magnitudes.

Test 2 - obvious real-world managers/funds

Without optimizing anything, inspect managers with clearly identifiable style mandates where available.

The objective is not to create a selected “smart money” universe. It is only a measurement sanity check.

Test 3 - holdings characteristic versus synthetic return beta

For each manager and factor, compare:

$$E_{mk,t}^{char} \quad \text{vs} \quad E_{mk,t}^{\beta}.$$

They should be related but not identical.

Large persistent discrepancies are informative and may signal:

- dynamic betas;
- industry effects;
- derivative hedges not visible in 13F;
- characteristic definitions that do not line up with the return factor.

Test 4 - factor portfolio reconstruction

Construct a known long/short factor portfolio and verify that its weighted stock characteristics have the expected sign and that the return series correlates strongly with the corresponding public factor return.

Test 5 - placebo randomization

Shuffle stock characteristics across stocks **within size/sector buckets**, recompute manager exposures, and show that economically meaningful cross-manager/time structure degrades.

This is stronger than a completely unrestricted shuffle because it preserves easy mechanical confounds.

18. Research hypotheses to pre-register

The current evidence suggests separating **state hypotheses** from **directional flow hypotheses**.

H1 - factor position is persistent

$$\text{Corr}(P_{k,t}, P_{k,t-1}) > 0.$$

This is primarily a measurement/state hypothesis.

H2 - active factor rotation is persistent over short horizons

$$\text{Rotation}_{k,t} \rightarrow \text{Rotation}_{k,t+1} > 0.$$

This is analogous to institutional trading persistence but at the factor level.

H3 - factor flow causes short-horizon continuation

$$\text{Pressure}_{k,t} > 0 \Rightarrow R_{k,t \rightarrow t+h} > 0$$

for a pre-registered short horizon.

Given the current project's evidence, this has a stronger prior than the opposite reversal hypothesis.

H4 - crowding alone has weak/no directional alpha

$$\text{CrowdingState}_{k,t} \not\Rightarrow \text{stable sign of future factor return.}$$

This is consistent with both the project's existing state-variable failures and mixed results in the crowding literature.

H5 - crowding amplifies adverse flow shocks

$$CrowdingState_{k,t} \times (-FundingFlow_{k,t})^+ \Rightarrow \text{larger negative factor return / tail risk.}$$

This is the preferred “crowding matters” specification.

H6 - factor rebalancing mismatch predicts stock-level pressure

Stocks whose current characteristics are misaligned with the persistent preferences of their institutional owners should experience subsequent owner-consistent trading pressure.

19. Statistical design with only about 50 independent quarters

The project sample is roughly 2013Q2-2026Q1, so a pure time-series factor test has very few independent quarterly observations.

This changes the test design.

Do not

- run dozens of factor definitions and report the best time-series Sharpe;
- treat every manager-stock row as an independent observation;
- infer strong significance from thousands of cross-sectional rows when the signal itself only changes quarterly.

Prefer

A. Panel tests across factors and time

For K factors and T quarters:

$$R_{k,t+1} = \alpha_k + \beta Pressure_{k,t} + \gamma Z_{k,t} + \epsilon_{k,t+1}.$$

Use time fixed effects or factor fixed effects where appropriate and standard errors clustered at least by quarter; two-way factor/time clustering is preferable when feasible.

B. Cross-sectional stock tests for factor-rebalancing extensions

This gives much richer variation while keeping the economic mechanism tied to 13F ownership.

C. Event-time tests around filing timestamps

If the hypothesis concerns disclosure/copycatting, use daily event windows and the actual acceptance timestamp rather than a quarterly portfolio-return regression.

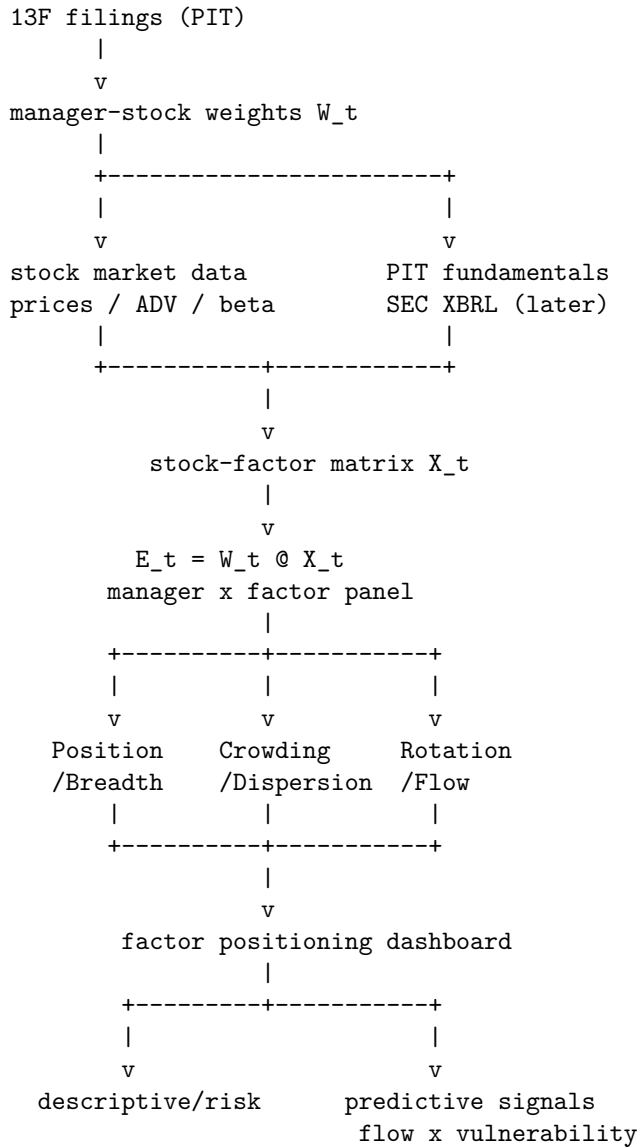
D. Block/bootstrap inference

Use time blocks at the quarter level for strategy statistics.

E. Pre-register the sign and horizon

Especially important because the project has already experienced economically meaningful signal inversions.

20. Recommended implementation architecture



Suggested repository layout

```

factors/manager_factor_positioning/
  README.md
  definitions.py
  build_stock_characteristics.py
  build_manager_exposures.py
  build_crowding_panel.py
  build_factor_rotation.py
  validate_exposures.py
  tests/

data/
  stock_factor_scores.parquet
  manager_factor_exposure.parquet
  factor_positioning_panel.parquet

reports/
  
```

```
measurement_validation.md
crowding_results.md
flow_results.md
```

21. Pseudocode for the first production version

```
# 1. Build point-in-time stock factor characteristics
X = build_characteristics(
    date=quarter_end,
    factors=["size", "momentum", "beta", "lowvol", "liquidity"],
    winsor=(0.01, 0.99),
    standardize="cross_sectional_zscore",
)

# 2. Add sector-neutral versions
X_sn = sector_neutralize(X, sector_map)

# 3. Build manager portfolio weights from 13F market values
W = holdings.pivot(index="filer_id", columns="security_id", values="weight")

# 4. Align universes and compute coverage
W, X = align(W, X)
coverage = W.notna_factor_coverage()

# 5. Manager x factor exposure
E_raw = W @ X
E_sn = W @ X_sn

# 6. Benchmark-relative exposure
benchmark_w = market_cap_weights(X.index)
benchmark_factor = benchmark_w @ X
E_active = E_raw - benchmark_factor

# 7. Aggregate positioning across managers
P_ew = E_active.mean(axis=0)
P_aum = weighted_mean(E_active, manager_13f_aum)

# 8. Breadth and dispersion
breadth = (E_active > 0.5).mean(axis=0)
dispersion = E_active.std(axis=0)

# 9. Store full manager-factor panel; do not collapse too early
save(E_raw, E_sn, E_active, coverage, metadata)
```

The critical rule is: **store the manager-level matrix permanently**. Do not only save the aggregate crowding index. Most future research ideas depend on the cross-section of manager exposures.

22. Priority roadmap

Priority 1 - Manager factor decomposition, market data only

Goal: create a stable $E_t = W_t X_t$ panel.

Implement:

- Size;
- Momentum;
- Beta;
- Low Volatility;
- Liquidity;
- sectors;
- raw and sector-neutral versions;
- active-vs-market benchmark versions.

Expected difficulty: low-medium.

Value: foundational. Almost every later positioning idea needs this table.

Priority 2 - Factor positioning dashboard

Implement:

- aggregate position;
- breadth;
- dispersion;
- positive/negative tail concentration;
- HHI of factor exposure across managers;
- quarter-on-quarter change.

Expected difficulty: low once E_t exists.

Do not trade this directly yet. First use it as descriptive state/risk.

Priority 3 - Factor rotation / factor-flow pressure

Implement the price-drift / characteristic-drift / trading decomposition and convert active trading into dollar factor flow.

This is the first genuinely predictive factor-level experiment I would prioritize.

Expected difficulty: medium.

Priority 4 - Funding-induced factor flow

Use the already validated implicit manager-flow instrument:

$$Flow_{k,t}^{funding} = \sum_m f_{m,t} E_{mk,t-1}.$$

Test whether factor price pressure is strongest when:

$$\text{funding shock} \times \text{factor crowding}$$

is large.

Expected difficulty: medium and highly synergistic with existing work.

Priority 5 - Value / Profitability / Investment via SEC XBRL

Only after the market-data block is stable.

Use filing timestamps to create true PIT fundamentals.

Expected difficulty: medium-high because accounting normalization is the real work.

Priority 6 - Factor rebalancing mismatch

Use persistent manager style preferences and stock characteristic drift to predict stock-level future holdings changes and returns.

Expected difficulty: medium-high.

Research appeal: very high.

Priority 7 - Factor Stress Matrix / eigenmodes

$$\Gamma_t = L_t^\top \Sigma_{u,t} L_t.$$

Use this as an advanced crowding/vulnerability product after the basic exposure and flow definitions have been validated.

Expected difficulty: high.

Research appeal: very high, but it should not precede the simple measurement layer.

23. What I would *not* implement first

Given the existing conclusions, I would deprioritize:

- another static PCA/NMF of holdings levels before classical factor exposure exists;
- another network centrality statistic without a validated directional shock;
- manager “skill” weights;
- filtering only to historically successful managers;
- a single scalar `factor_crowding_score` optimized for Sharpe;
- full Barra-style estimation before the simpler characteristic layer is working;
- dozens of accounting factors sourced from non-PIT Yahoo data.

The simple classical decomposition is useful precisely because it gives an interpretable coordinate system for the more sophisticated flow and network signals already developed.

24. Literature review - papers most relevant to this implementation

24.1 Daniel, Grinblatt, Titman and Wermers (1997) - characteristic-based benchmarks

Paper: *Measuring Mutual Fund Performance with Characteristic-Based Benchmarks*, Journal of Finance 52(3), 1035-1058.

Why it matters here: establishes the classic holdings/characteristics perspective for evaluating portfolio style and performance. It is a strong conceptual foundation for representing a portfolio by the characteristics of the stocks it actually owns instead of relying only on return regressions.

Implementation takeaway: a holdings snapshot can be mapped into interpretable style coordinates directly.

24.2 Lettau, Ludvigson and Manoel (2018/2021) - characteristics of fund portfolios

Paper: *Characteristics of Mutual Fund Portfolios: Where Are the Value Funds?*, NBER Working Paper 25381.

The paper studies the actual characteristics of mutual fund, ETF and hedge-fund portfolios and shows that stated fund labels need not map cleanly to extreme characteristic tilts. It reports especially strong concentration of funds in large stocks, while some other characteristic distributions are much closer to the market.

Implementation takeaway: use the holdings themselves to classify style, and benchmark exposures relative to the market so that mechanical large-cap bias is not mistaken for an informative factor bet.

24.3 Grinblatt, Jostova, Petrasek and Philipov (2020) - 13F reveals investment style

Paper: *Style and Skill: Hedge Funds, Mutual Funds, and Momentum*, Management Science 66(12), 5505-5531.

The study uses mandatory institutional 13F holdings to identify differences in trading style across hedge funds and mutual funds, particularly momentum versus contrarian behavior.

Implementation takeaway: 13F holdings are informative enough to classify persistent manager style and trading orientation, even though they do not contain the complete hedge-fund portfolio.

24.4 Sias (2004) - institutional demand persistence and herding

Paper: *Institutional Herding*, Review of Financial Studies 17(1), 165-206.

Institutional demand for a security is positively related to lagged institutional demand, with evidence of institutions following both other institutions and their own prior trades.

Implementation takeaway: after constructing factor-level holdings, test whether institutional demand persistence also exists in **factor space**, not only stock space.

24.5 Brown, Howard and Lundblad (2022) - crowded trades and tail risk

Paper: *Crowded Trades and Tail Risk*, Review of Financial Studies 35(7), 3231-3271. DOI 10.1093/rfs/hhab107.

The paper constructs security-level crowdedness measures from hedge-fund holdings and finds that exposure to crowdedness helps explain downside tail risk during industry distress.

Implementation takeaway: crowding can be relevant as a vulnerability dimension. It does not imply that a static crowded factor should mechanically be shorted.

24.6 Barroso, Edelen and Karehnke (2022) - caution against crowding as stand-alone crash signal

Paper: *Crowding and Tail Risk in Momentum Returns*, Journal of Financial and Quantitative Analysis 57(4), 1313-1342. DOI 10.1017/S0022109021000624.

The paper finds that institutional-holdings crowding proxies do not support a simple universal story that greater crowding mechanically means greater expected momentum crash risk.

Implementation takeaway: this reinforces the current project's own result: use crowding mainly as state/vulnerability and add a directional flow shock.

24.7 Ben-David, Li, Rossi and Song (2024) - institutional demand can move factor/style returns

Paper: *Discontinued Positive Feedback Trading and the Decline of Return Predictability*, Journal of Financial and Quantitative Analysis 59, 3062-3100; earlier NBER Working Paper 28624.

The paper links an institutional change in fund demand to changes in systematic return-predictability patterns and uses holdings-based measures of style exposure.

Implementation takeaway: factor-level institutional demand is not merely descriptive; coordinated demand for styles can affect factor returns.

24.8 Dou, Kogan and Wu (2022; Journal of Finance forthcoming/updated working paper) - common fund flows and factor pricing

Paper: *Common Fund Flows: Flow Hedging and Factor Pricing*, NBER Working Paper 30234.

The authors model and empirically study common fund-flow shocks and managers' portfolio responses to those shocks.

Implementation takeaway: the covariance structure of manager funding shocks is economically relevant and motivates the proposed factor stress matrix Γ_t .

24.9 Peng and Wang (2026) - factor rebalancing

Paper: *Factor Rebalancing*, working paper, SSRN abstract 3327849, 2026 version.

The paper documents that mutual funds with persistent factor demand rebalance as stock characteristics drift. Focusing on value and momentum, it reports predictable price pressure associated with stocks becoming misaligned with the factor preferences of their owners.

Implementation takeaway: this is perhaps the most direct literature bridge from a manager-factor exposure matrix to a new stock-level predictive 13F signal.

24.10 Fama-French and Kenneth French Data Library - canonical factor construction

For canonical terminology and portfolio construction, the Kenneth French Data Library remains the clean reference point:

- SMB: size;
- HML: book-to-market/value;
- RMW: profitability;
- CMA: investment;
- Momentum: size x prior 2-12 month return portfolios.

Implementation takeaway: use canonical sign conventions and factor definitions where practical, but remember that a **stock characteristic score** and a **factor portfolio return beta** are distinct objects.

25. Interpretation: what a senior quant should expect from the output

The goal is not merely to produce a chart saying “Momentum is crowded.”

The useful end-state is a layered answer.

For example:

Factor: Momentum

13F long-equity aggregate tilt:	+1.5 z
Breadth of positive-tilt managers:	71%
Exposure dispersion:	low
Positive-tail concentration:	medium
Active rotation this quarter:	+0.8 z
Funding-induced flow:	-1.2 z
Funding fragility:	+1.7 z
Dominant stress-mode loading:	high

Economic interpretation:

Momentum is structurally crowded and broadly held. Managers continued to rotate into it on the last reported quarter, but the current funding shock points in the opposite direction and is concentrated among managers whose flows historically co-move. This is not merely “crowded Momentum”; it is a potentially vulnerable crowded factor experiencing an adverse capital shock.

That is substantially more useful than a single static factor exposure statistic.

26. Final recommendation

The manager-factor decomposition should be treated as **infrastructure**, not as one more isolated alpha test.

Build first:

$$E_t = W_t X_t$$

with a small, clean set of market-data factors.

Then add three layers in order:

$$\text{Positioning} \rightarrow \text{Factor Rotation} / \text{Flow} \rightarrow \text{Flow} \times \text{Crowding Vulnerability}$$

The most promising immediate predictive test is **factor-flow pressure**, not static factor crowding.

The most promising later stock-level extension is **factor rebalancing mismatch**.

The most interesting mathematical product is the factor stress matrix:

$$\Gamma_t = L_t^\top \Sigma_{u,t} L_t$$

but it should only be built after the simpler manager-factor exposure panel has passed basic measurement validation.

If this research sequence works, the final project gains a clear hierarchy:

1. **Stock-level 13F alpha:** fresh conviction and distress flows;
2. **Manager-level state:** classical factor exposure of each long-equity book;
3. **Market-level positioning:** how institutional capital is distributed across factors;
4. **Directional factor flow:** where that capital is moving;
5. **Systemic vulnerability:** which factor combinations are held by capital exposed to common funding shocks.

That is a coherent factor-positioning framework rather than another isolated 13F signal.

References

- Barberis, N. and Shleifer, A. (2003). *Style Investing*. Journal of Financial Economics 68, 161-199. Earlier NBER Working Paper 8039.
- Barroso, P., Edelen, R. M., and Karehnke, P. (2022). *Crowding and Tail Risk in Momentum Returns*. Journal of Financial and Quantitative Analysis 57(4), 1313-1342. DOI: 10.1017/S0022109021000624.
- Ben-David, I., Li, J., Rossi, A., and Song, Y. (2024). *Discontinued Positive Feedback Trading and the Decline of Return Predictability*. Journal of Financial and Quantitative Analysis 59, 3062-3100. Earlier NBER Working Paper 28624.
- Brown, G. W., Howard, P., and Lundblad, C. T. (2022). *Crowded Trades and Tail Risk*. Review of Financial Studies 35(7), 3231-3271. DOI: 10.1093/rfs/hhab107.
- Daniel, K., Grinblatt, M., Titman, S., and Wermers, R. (1997). *Measuring Mutual Fund Performance with Characteristic-Based Benchmarks*. Journal of Finance 52(3), 1035-1058.
- Dou, W. W., Kogan, L., and Wu, W. (2022). *Common Fund Flows: Flow Hedging and Factor Pricing*. NBER Working Paper 30234; revised versions circulated subsequently.
- Fama, E. F. and French, K. R. Factor definitions and portfolio construction, Kenneth French Data Library.
- Grinblatt, M., Jostova, G., Petrsek, L., and Philipov, A. (2020). *Style and Skill: Hedge Funds, Mutual Funds, and Momentum*. Management Science 66(12), 5505-5531. DOI: 10.1287/mnsc.2019.3433.
- Greenwood, R. and Thesmar, D. (2011). *Stock Price Fragility*. Journal of Financial Economics 102(3), 471-490.
- Lettau, M., Ludvigson, S. C., and Manoel, P. (2018). *Characteristics of Mutual Fund Portfolios: Where Are the Value Funds?* NBER Working Paper 25381.
- Peng, C. and Wang, C. (2026). *Factor Rebalancing*. Working paper, SSRN abstract 3327849.
- Sias, R. W. (2004). *Institutional Herding*. Review of Financial Studies 17(1), 165-206.
- U.S. Securities and Exchange Commission. *Frequently Asked Questions About Form 13F* and Form 13F data documentation.